Assignment 3: Demon Slayer Corps, The String Hunt

Roll No: 24I-3060 | Section: SE-A | Language Chosen: Python

Final Selection: test_matchers.py

All four matchers (naive, Rabin-Karp, KMP, automaton) plus the prefix function and KMP chunk-resume pass every test.

```
PS C:\Users\User\Documents\GitHub\DemonOfSongHunt\starter> python test_matchers.py
PASS naive       aba in ababa
PASS rabin-karp  aba in ababa
PASS kmp         aba in ababa
PASS automaton   aba in ababa
PASS naive       ababaca in abababacaba
PASS rabin-karp  ababaca in abababacaba
PASS kmp         ababaca in abababacaba
PASS automaton   ababaca in abababacaba
PASS naive       aa in aaaaaa
PASS rabin-karp  aa in aaaaaa
PASS kmp         aa in aaaaaa
PASS automaton   aa in aaaaaa
PASS naive       abcab in acaabcaabcab
PASS rabin-karp  abcab in acaabcaabcab
PASS kmp         abcab in acaabcaabcab
PASS automaton   abcab in acaabcaabcab
PASS naive       d in abcabcabc
PASS rabin-karp  d in abcabcabc
PASS kmp         d in abcabcabc
PASS automaton   d in abcabcabc
PASS prefix function of ababaca
PASS rabin-karp shifts with d=10, q=11
PASS rabin-karp spurious hits with d=10, q=11
PASS kmp resume across a chunk boundary
```

Mission 1: The Crow's Alarm (KMP)

KMP reads crow_log.txt in 4 KB chunks. The state q is carried from one chunk into the next, so a KAKAKA split across a chunk boundary is still counted.

```
PS C:\Users\User\Documents\GitHub\DemonOfSongHunt\starter> python mission1.py
pi = [0, 0, 1, 2, 3, 4]
Alarm count: 415
First 10 shifts: [8620, 8622, 8624, 39118, 39126, 88124, 88132, 90213, 90215, 90217]
```

Mission 2: Tamayo's Blood Lab (finite automaton)

A delta table is built for GATTACAGATTC and for its reverse complement GAATCTGTAATC. Both automata run on each sample, and the demon is the sample with marker hits.

```
PS C:\Users\User\Documents\GitHub\DemonOfSongHunt\starter> python mission2.py
delta for GATTACAGATTC
state  A  C  G  T
    0  0  0  1  0
    1  2  0  1  0
    2  0  0  1  3
    3  0  0  1  4
    4  5  0  1  0
    5  0  6  1  0
    6  7  0  1  0
    7  0  0  8  0
    8  9  0  1  0
    9  0  0  1  10
   10  0  0  1  11
   11  5  12  1  0
   12  0  0  1  0

reverse complement of the marker: GAATCTGTAATC

file           forward  reverse
sample_01.txt        0        0
sample_02.txt        0        0
sample_03.txt        0        0
sample_04.txt        0        0
sample_05.txt        0        0
sample_06.txt        0        0
sample_07.txt        0        3
sample_08.txt        0        0
sample_09.txt        0        0
sample_10.txt        0        0

The demon is: sample_07.txt (forward 0, reverse 3)
```

Mission 3: The Copycat Slayer (Rabin-Karp)

Every text is cleaned (lowercase, punctuation removed, single spaces). Every 40-character window of the manual goes into a hash map (d = 256, q = 1,000,000,007). The hash is rolled across each report, and each hit is verified character by character.

```
PS C:\Users\User\Documents\GitHub\DemonOfSongHunt\starter> python mission3.py
similarity per trainee
genya        0.00%
inosuke     14.96%
kanao       31.86%
tanjiro      0.00%
zenitsu     89.00%

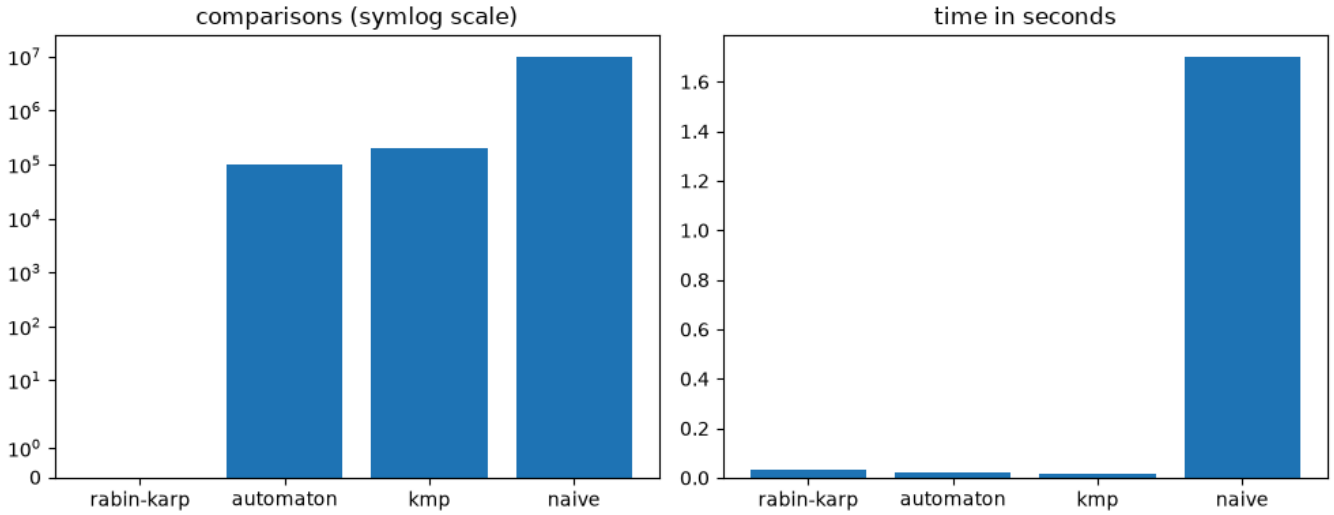
ranking (highest first)
1. zenitsu     89.00%
2. kanao       31.86%
3. inosuke     14.96%
4. tanjiro      0.00%
5. genya        0.00%
```

Why the trainee who changed words scores lower than deserved:

The detector only counts text as copied when a full 40-character window matches the manual exactly. Swapping words breaks almost every window, and the unchanged stretches between edits are too short to fill one, so most of their real copying goes uncounted.

Bonus: Breathing Styles Tournament

Arena: T = 'a' x 100,000 and P = 'a' x 99 + 'b'. All four techniques race on the same input.



Results: comparisons / time

Rabin-Karp: 0 / 0.033 s. Automaton: 100,000 / 0.022 s. KMP: 199,901 / 0.015 s. Naive: 9,990,100 / 1.70 s.

Winner and loser:

Loser, **naive**: at each of the 99,901 shifts it matches 99 a's and fails on the b, so it makes 100 comparisons per shift and about 10 million in total.

KMP and the automaton never re-read text: about 2n comparisons for KMP and exactly n table lookups for the automaton, so both stay linear.

Winner, **Rabin-Karp**: no window hash equalled the hash of P, so it never had to verify anything and made 0 comparisons. This depends on the hash values and not the algorithm, since a collision would force a full check. In time, the three non-naive methods all finish within about 0.03 s, so the gap to naive (1.70 s) is what matters.